

Magnetic field of a magnetic loop antenna for COHIRADIA applications

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1 THEORY

When building a loop antenna, the question arises as to the optimal choice of parameters—such as dimensions and number of turns—if the goal is to generate the strongest possible magnetic field at a certain distance. In doing so, certain constraints must be taken into account, such as the supply voltage of the driver amplifier, its maximum deliverable current, and the maximum allowable power dissipation within the linear operating range.

To model this, we start with a circular conductor loop that is assumed to be “thin,” i.e., the diameter d of the conductor bundle is small compared to the radius of the loop.

As described, for example, at [this link](#), the axial component B_z of the magnetic field of such a current-carrying circular loop is calculated as

$$B_z = \frac{\mu_0 I a^2}{2(a^2 + z^2)^{\frac{3}{2}}} \quad 1-1$$

with a = loop radius, I = loop current and z = distance from the loop center on the axis. μ_0 is the permeability constant of free space. If N turns are used, and assuming that the diameter of the conductor bundle is small compared to the loop radius (i.e., not a solenoid coil), the following approximation holds:

$$B_z = N \frac{\mu_0 I a^2}{2(a^2 + z^2)^{\frac{3}{2}}} \quad 1-2$$

The inductance is calculated ([see e.. this online calculator](#)) as:

$$L = \mu_0 a N^2 \left(\ln \left(\frac{4a}{d} \right) - 2 \right) \quad 1-3$$

where d is the wire diameter. Suppose we connect this coil in series with a resistor R (including the coil's loss resistance) to serve as the amplifier's load impedance Z_L ; then we obtain:

$$Z_L = R + j\omega L = R + j\omega \mu_0 a N^2 \left(\ln \left(\frac{4a}{d} \right) - 2 \right) \quad 1-4$$

The rms current through the coil, given the rms voltage V_L at the load at radian frequency ω is:

$$I = \frac{V_L}{R + j\omega L} = \frac{V_L}{R + j\omega \mu_0 a N^2 \left(\ln \left(\frac{4a}{d} \right) - 2 \right)} \quad 1-5$$

The boundary conditions are defined by the amplifier's maximum deliverable current I_{max} , the maximum deliverable output voltage amplitude V_L , and, if applicable, the maximum allowable power dissipation in the amplifier P_{max} .

Power dissipation of the amplifier: The power dissipation in the amplifier is the difference between the power drawn from the power supply, P_S , and the power delivered to the load, P_L . For sinusoidal excitation and an ohmic-inductive load, given a (bipolar) supply voltage V_S , the following can be written (where V_S is given as the rms value):

$$P_D = P_S - P_L = V_S I_L - \text{Re}(V_L I_L^*) \quad 1-6$$

$$P_D = \frac{V_S V_L}{|Z|} - \text{Re}\left(\left(\frac{V_S}{Z}\right)^* V_S\right) = \frac{V_S V_L}{\sqrt{R^2 + (\omega L)^2}} - \text{Re}\left(\frac{V_L^2}{R - j\omega L}\right) \quad 1-7$$

Expansion of the second term yields::

$$P_D = \frac{V_S V_L}{\sqrt{R^2 + (\omega L)^2}} - \text{Re}\left(\frac{V_L^2 (R + j\omega L)}{R^2 + (\omega L)^2}\right) = \quad 1-8$$

$$\frac{V_S V_L}{\sqrt{R^2 + (\omega L)^2}} - \frac{V_L^2 R}{R^2 + (\omega L)^2}$$

Simple check: In case of purely ohmic load ($L = 0$) we obtain (as expected):

$$P_D = \frac{V_S V_L}{R} - \frac{V_L^2}{R} = (V_S - V_L) \frac{V_L}{R} \quad 1-9$$

With minimal power dissipation at full modulation, i.e. $V_L = V_S/\sqrt{2}$, we have:

$$P_D = \frac{V_S^2}{R} \left(1 - \frac{1}{\sqrt{2}}\right) \frac{1}{\sqrt{2}} = \frac{V_S^2}{R} \frac{\sqrt{2}-1}{2} = 0.205 \frac{V_S^2}{R}$$

However, in broadband operation, a highly inductive load is of greater practical significance. **For a purely inductive load** ($L = 0$), the power dissipation is:

$$P_D = \frac{V_S V_L}{\omega L} \quad 1-10$$

This reaches its maximum value at full modulation ($V_L = V_S/\sqrt{2}$), in contrast to the ohmic case. This also determines the **maximum** permissible current at full modulation:

$$|I_{max}| = \min\left(I_{lim}, \frac{V_L}{\omega L}\right) = \min\left(I_{lim}, \frac{P_D}{V_S}\right)$$

i.e., limited either by the maximum power dissipation or the maximum deliverable output current, whichever is lower.

The maximum field is thus automatically determined for the number of turns N at which the maximum permissible current can flow at a given supply voltage:

$$I_{\max} = \frac{V_L}{j\omega\mu_0 a N^2 \left(\ln\left(\frac{4a}{d}\right) - 2 \right)} \quad 1-11$$

With eq 1-2 the maximum rms magnetic flux density can be obtained as:

$$B_z = \frac{V_s a}{2\sqrt{2} j\omega N (a^2 + z^2)^{\frac{3}{2}} \left(\ln\left(\frac{4a}{d}\right) - 2 \right)} \quad 1-12$$

For $z = 0$:

$$B_z \approx \frac{V_s}{2\sqrt{2} j\omega N a^2 \left(\ln\left(\frac{4a}{d}\right) - 2 \right)} \quad 1-13$$

If $z \gg a$ we get approximately:

$$B_z \approx \frac{V_s a}{2\sqrt{2} j\omega N z^3 \left(\ln\left(\frac{4a}{d}\right) - 2 \right)} \quad 1-14$$

2 PRACTICAL APPLICATION

This immediately shows that the field decreases as the number of turns increases and increases as the radius increases, though not proportionally. Large coils with few turns therefore perform better at long distances than small coils with many turns. Fig. 1 shows the generated field (in T_{RMS}) as a function of distance z and number of turns N at a radius of 10 cm for a unipolar supply voltage of 9 V (i.e., $V_s = 4.5$ V). The allowable power dissipation was assumed to be 1 W and the maximum deliverable current 1 A. On the left is an isoline plot and on the right a heatmap with logarithmic scaling; the information is essentially the same. The zone where the maximum current is exceeded (here due to the power dissipation limit) is marked in red.

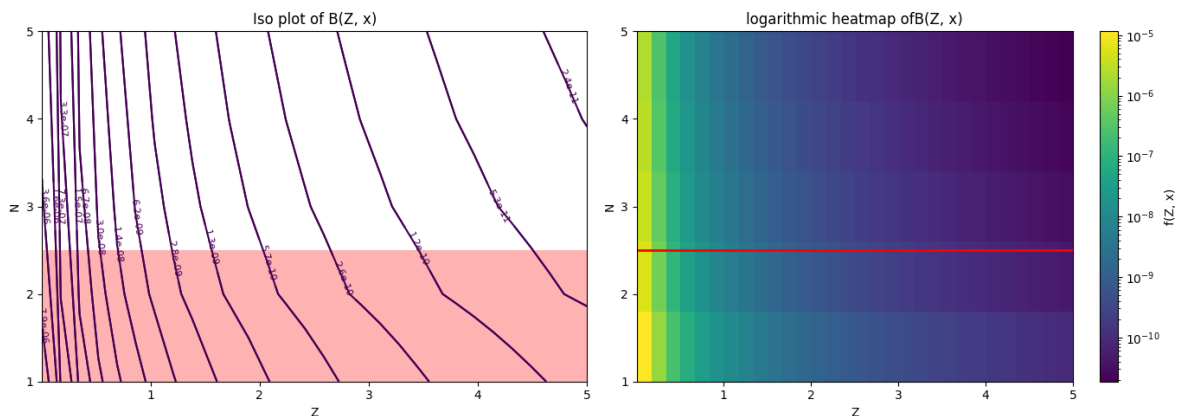


Fig. 1: The z-component of the magnetic flux density (T) as a function of the axial distance from the transmitter coil and the number of turns N , with a loop radius of 0.1 m. The heatmap is logarithmic for better visualization of the low flux density regions!

As can be seen, although the low inductance at low numbers of turns results in the highest possible magnetic flux densities, it also leads to unacceptably high currents. Therefore, one must either use more than one turn or connect a resistor in series with the coil. If the radius is doubled (20 cm), the situation improves, as shown in Fig. 2

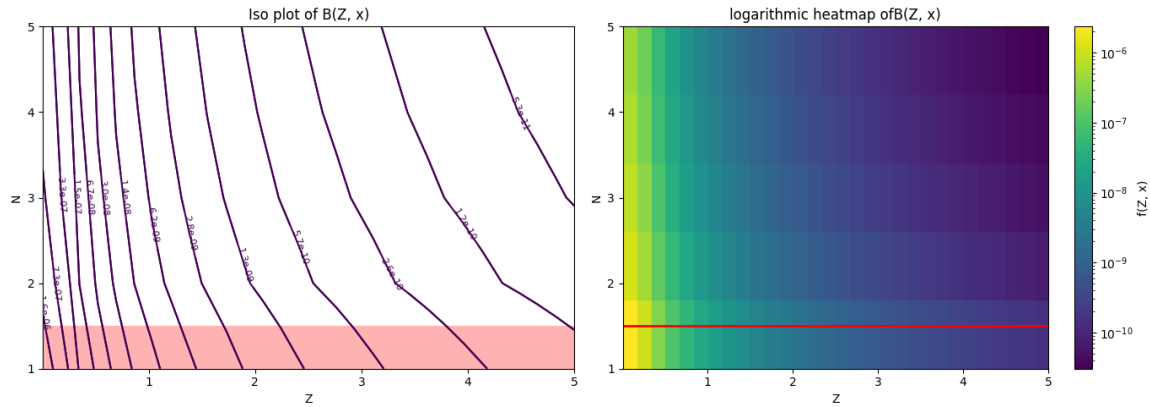


Fig. 2: The z -component of the magnetic flux density (T) as a function of the axial distance from the transmitter coil and the number of turns N , with a loop radius of 0.2 m. The heatmap is logarithmic for better visualization of the low flux density regions!

Small-signal amplifiers such as the AD811 typically allow a maximum of 70 mA rms current, which requires a resistor of, say, 50 Ω to be connected in series. As Fig. 3 shows, the field is then weaker, but is usually sufficient to power a portable radio over a distance of 2–3 m. In this illustration, up to 50 turns were used to clearly demonstrate the effect of N . For longer distances, 2–3 turns are slightly more favorable than a single turn; beyond that, the situation deteriorates. Near the transmitter coil, the field drop with N is not as severe, since significantly higher flux densities are present there anyway, but at greater distances, high numbers of turns have a negative effect.

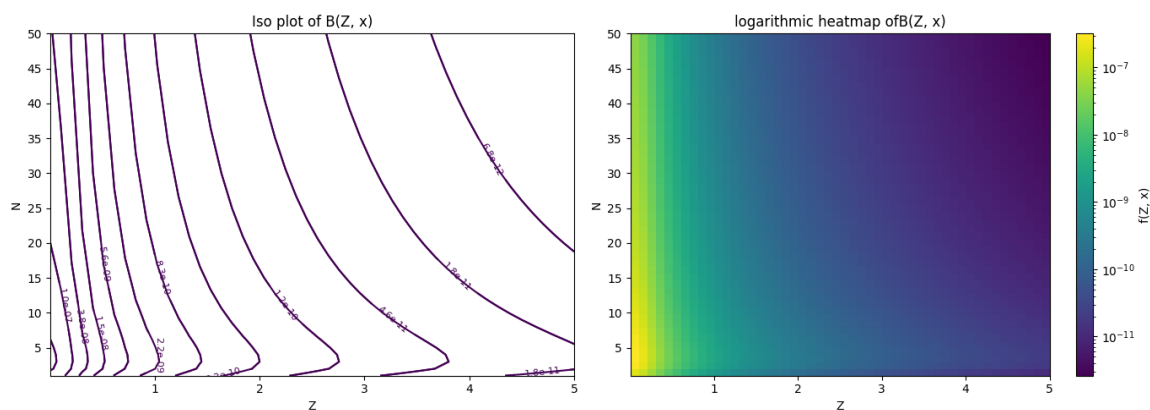


Fig. 3: z -component of the magnetic flux density (T) as a function of the axial distance from the transmitter coil and the number of turns N , with a loop radius of 0.2 m and a maximum effective loop current of 70 mA_{eff}. This can be achieved using an AD811 with a series resistor of 50 Ω . The heat map is logarithmic for better visibility of the small flux density ranges!

